RESULTS AND OBSERVATION

The proposed system adopts a hybrid classification approach based on a stacking ensemble technique as shown in Fig 1.

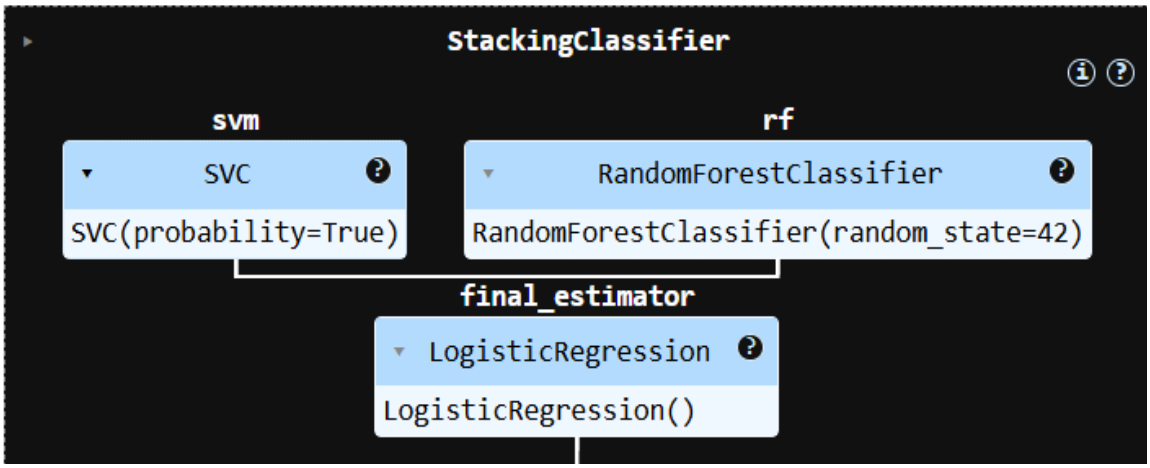


Fig 1 - Stacking-Based Hybrid Model Architecture for Thermal Fault Detection

It is clear from Fig 1 that Support Vector Classifier (SVC) and Random Forest (RF) are used as base classifiers for this case. These models represent different characteristics of the data, such as linear and non-linear decision boundaries. Their outputs are then given to a Logistic Regression model, which serves as the meta-learner. This stacking-based hybrid approach improves the classification performance.

1 PERFORMANCE ANALYSIS OF LBP-BASED MODEL

To establish a baseline for comparison, the performance of the model using Local Binary Pattern (LBP) features is evaluated. This helps in understanding the effectiveness of traditional texture-based feature extraction for thermal fault detection.

The classification report and confusion matrix obtained using the LBP approach is shown in Fig 2.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        64
           1       0.64      1.00      0.78       116

    accuracy                           0.64       180
   macro avg       0.32      0.50      0.39       180
weighted avg       0.42      0.64      0.51       180
```

Fig 2 - Classification Report for LBP-based Model

From Fig 2, it is observed that the model shows poor performance in classifying class 0 (Normal Condition), with precision, recall, and F1-score values close to zero. This indicates that the model is unable to correctly identify normal samples. In contrast, for class 1 (Abnormal Condition), the model achieves a higher recall value, suggesting that most abnormal samples are classified correctly.

The confusion matrix in Fig 3 further highlights this behavior, where a majority of the samples are classified as class 1. Specifically, all normal samples are misclassified as abnormal, resulting in the absence of true negatives. This leads to a strong bias toward the abnormal class.

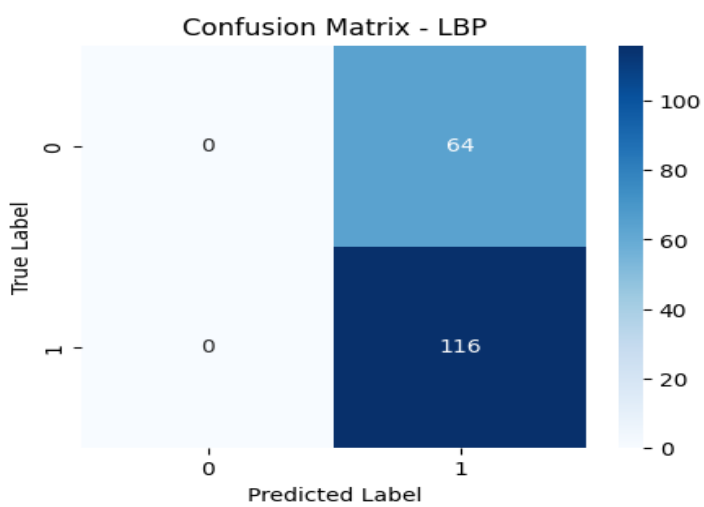


Fig 3 - Confusion Matrix for LBP-based Model

Although the overall accuracy appears moderate, the macro and weighted average scores indicate poor class balance and unreliable performance.

2 PERFORMANCE ANALYSIS OF EFFICIENT NET- BASED MODEL

The EfficientNet-based model is evaluated to understand how effectively it learns and interprets complex thermal patterns from the dataset. When compared to the conventional

LBP-based approach, the results show a notable improvement across all key performance indicators.

The classification report obtained using the EfficientNet model is represented in Fig The generated classification report proves that the model achieves significantly improved performance compared to the LBP-based approach. For the case of Class 0, which corresponds to normal operating conditions,the precision and recall values are both high, indicating accurate identification of normal samples. Similarly, for class 1, corresponding to the abnormal case, and it showcases the model performance with high recall and F1-score, and detect fault conditions effectively.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.92      0.86      0.89        64
           1       0.93      0.96      0.94       116

    accuracy                           0.92       180
   macro avg       0.92      0.91      0.91       180
weighted avg       0.92      0.92      0.92       180
```

Fig 4 - Classification Report for EfficientNet-B0 Model

The confusion matrix in Fig 5 further supports these observations. A majority of samples are correctly classified, with 55 cases being true negative and 111 instances true positives. There are only a few cases of error in classification, which include a few false positives and false negatives.

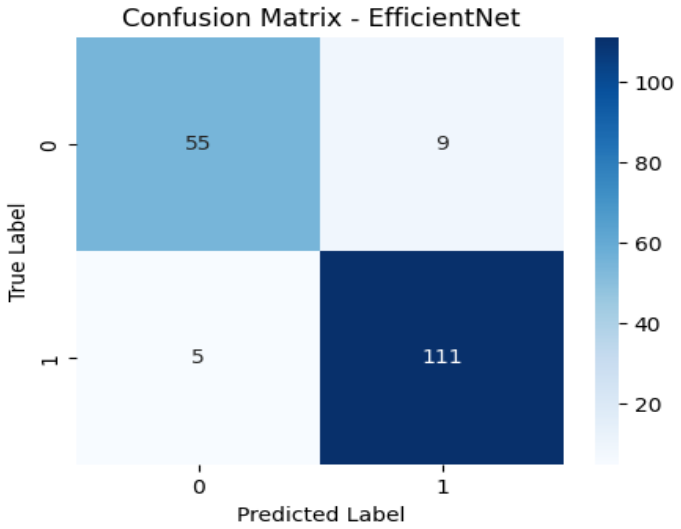


Fig 5 - Confusion Matrix for EfficientNet-B0 Model

The overall accuracy of the model is high, and both macro and weighted averages reflect consistent performance across classes, indicating good generalization capability.

3 PERFORMANCE ANALYSIS OF HYBRID MODEL

The performance of the hybrid model is analyzed to evaluate its effectiveness of combining texture-based features with deep learning for thermal fault detection.

The classification report generated using the hybrid approach is shown in Fig 6.The results show that there is a significant improvement made by the hybrid method compared to just using the EfficientNet approach.

	precision	recall	f1-score	support
0	0.92	0.89	0.90	64
1	0.94	0.96	0.95	116
accuracy			0.93	180
macro avg	0.93	0.92	0.93	180
weighted avg	0.93	0.93	0.93	180

Fig 6 - Classification Report for Hybrid Model

The number of correct classifications is indicated in Fig 7 as most of the samples are identified accurately, where there are 57 true negative and 11 true positive values. However, it has been observed that the errors in classification are also very small in comparison to the EfficientNet model.

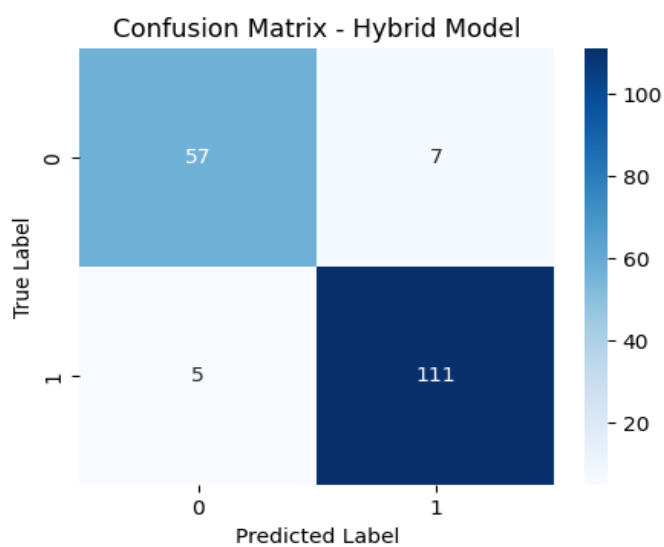


Fig 7 - Confusion Matrix for Hybrid Model

Overall, the accuracy of the proposed hybrid model is higher, and from the macro and weighted averages, we get to know about the balanced and consistent performance across both classes. It suggests that the integration of LBP features with EfficientNet enhances both feature representation and classification capability.

4 COMPARATIVE ANALYSIS OF MODELS

We perform a comparative evaluation to assess the performance of the LBP, EfficientNet and Hybrid models. Table 1 presents the accuracy comparison among the three approaches.

Table 1 - Accuracy Comparison of LBP, EfficientNet, and Hybrid Models

Model	Accuracy
LBP	64.44%
EfficientNet	92.22%
Hybrid (LBP + EfficientNet)	93.33%

Table 1 shows the LBP model records the lowest accuracy at around 6 44%, indicating its limitation in capturing complex thermal patterns. The EfficientNet model significantly improves performance, achieving an accuracy of approximately 92.22%, demonstrating the strength of deep learning in feature extraction.

The Hybrid model achieves the highest accuracy of approximately 93.33%, slightly outperforming the EfficientNet model. The bar chart clearly highlights this improvement, confirming the benefit of combining LBP features with deep learning.The hybrid model delivers the best performance, showing that integrating traditional and deep learning features enhances accuracy and robustness.

5 MODEL INTERPRETABILITY AND EXPLANATION ANALYSIS

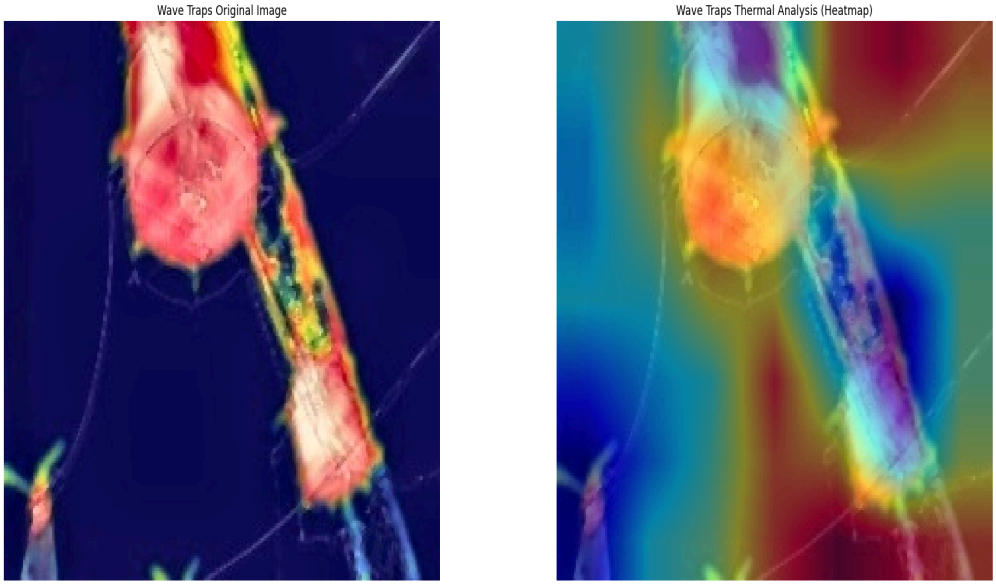
An analysis of the model's decision-making process is essential to understand whether the predictions are based on meaningful thermal features. In this context, XAI techniques are applied to interpret the behavior of the hybrid model.

In the proposed approach, Grad-CAM is used to identify the spatial regions within the thermal image that influence the classification output, particularly for detecting abnormal conditions. It highlights areas of high thermal intensity, enabling localization of potential fault regions. SHAP, on the other hand, is used to quantify the contribution of different regions of the image toward the final prediction, indicating how strongly each region influences the classification decision. In addition to these techniques, a lightweight language model, TinyLLaMA, is integrated to generate descriptive explanations based on the model predictions and thermal analysis. This enables the system to provide not only visual interpretations but also contextual insights regarding the possible cause and operational implications of the detected condition.

5.1 Abnormal Condition - Wave Traps

The original thermal image, the heatmap, and the feature contribution plot is shown in Fig 8. The image illustrates the presence of a hotspot in the upper and lower section of the

Wavetrap, which denotes a highly concentrated thermal area that corresponds to an anomaly. The Grad-CAM heatmap shown in Fig 8 locates the area as the lower section and classifies the object as abnormal. The SHAP analysis supports this by providing higher importance to the same region, while minimising the contribution to other areas.




```
Equipment: Wave Traps
Condition: Abnormal
Hotspot Location: lower section
Fault Probability: 0.9

AI Explanation:
 Electrical Cause: High voltage (HV) leakage current within the wire
insulation
Operational Risk: Electrical shock hazard to personnel working in close
proximity to the circuit breaker or switchgear control panel.

The HV leakage currents generated by the faulty equipment were caused
due to high voltage (HV) leakage current within the wire insulation.
This condition increases with prolonged exposure to high voltage,
leading to thermal conditions that can lead to high-voltage (HV) leaks
or arc flashes. These leaks increase the chances of sparking or arcing
between two conductors of the circuit and potentially causing electric
shocks or burn injuries to personnel nearby. The incident had resulted
from improper maintenance procedures being carried out at the site
without proper safety precautions.
```

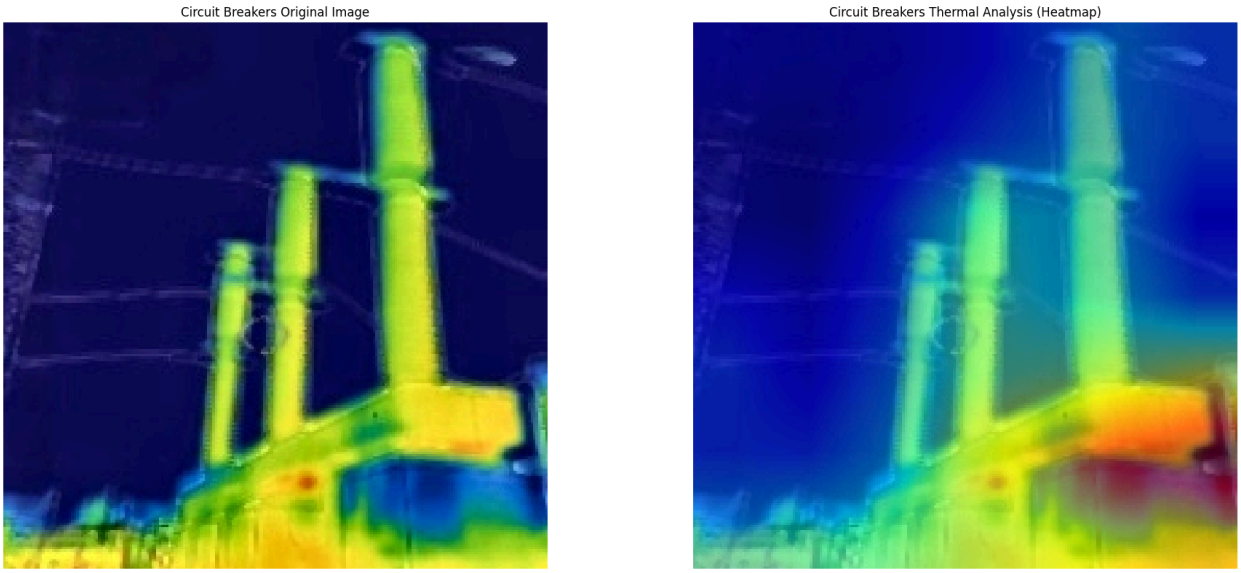
Fig 8 - Grad-CAM and SHAP-Based Interpretation of Abnormal Condition in Wave Trap with AI-Based Explanation

The high fault probability shown in Fig 8 reflects the severity of the detected anomaly and corresponds to potential high voltage leakage within the insulation.

5.2 Normal Condition - Circuit Breakers

The original thermal image, its corresponding heatmap, and the feature contribution analysis shown in Fig. 9 provides a result that the Circuit Breaker is in Normal operating

condition. However, a very minor thermal gradient could be observed in the lower section, whose severity is very low and therefore supports the classification of the model. Grad-CAM produces diffuse activation across the image, thereby indicating the absence of dominant fault regions. SHAP analysis also shows low and evenly distributed feature importance, confirming that no specific region strongly influences the prediction.




```
Equipment: Circuit Breakers
Condition: Normal
Hotspot Location: lower section
Fault Probability: 0.1

AI Explanation:
 The circuit breakers are designed to be used in a temperature range
that is within their normal operating limits for safe and reliable
performance. When the circuit breaker reaches its hot spot or limit
switch position, it automatically disconnects the circuit to prevent
damage to the electrical system. This helps protect against overload
conditions and minimizes hazards associated with high temperatures. The
thermal condition is normal as indicated by the location where the
hotspots occur - typically located at the bottom of the device.
```

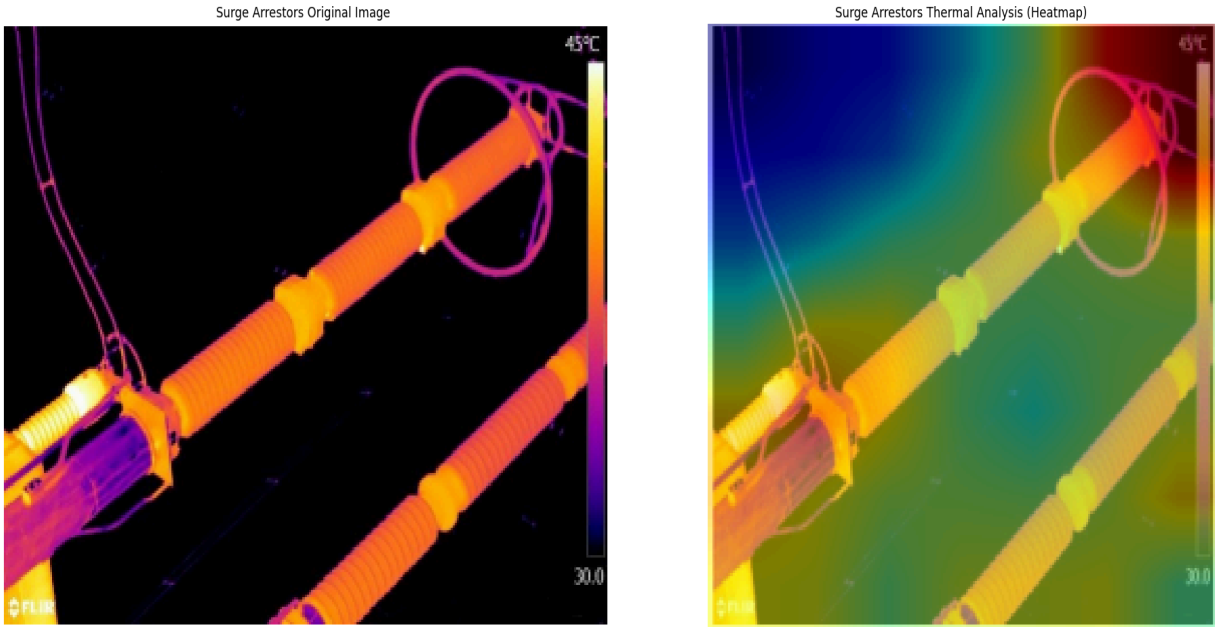
Fig 9 - Grad-CAM and SHAP-Based Interpretation of Normal Condition in Circuit Breaker with AI-based explanation

The low fault probability shown in Fig 9 indicates that the equipment is operating within safe thermal limits.

5.3 Abnormal Condition - Surge Arrestors

The visualization in Fig. 10, comprising the original thermal image, its corresponding heatmap, and feature contribution analysis, depicts a hotspot located in the upper section of the

surge arrester. The thermal image reveals an abnormal heat buildup in a localized region, where the temperature is visibly elevated compared to the surrounding areas. The Grad-CAM heatmap points directly to this upper section as the key region driving the fault classification, confirming that the model has correctly pinpointed the problem area. SHAP analysis backs this up by assigning greater importance to the same region, while the rest of the image contributes very little to the final prediction.



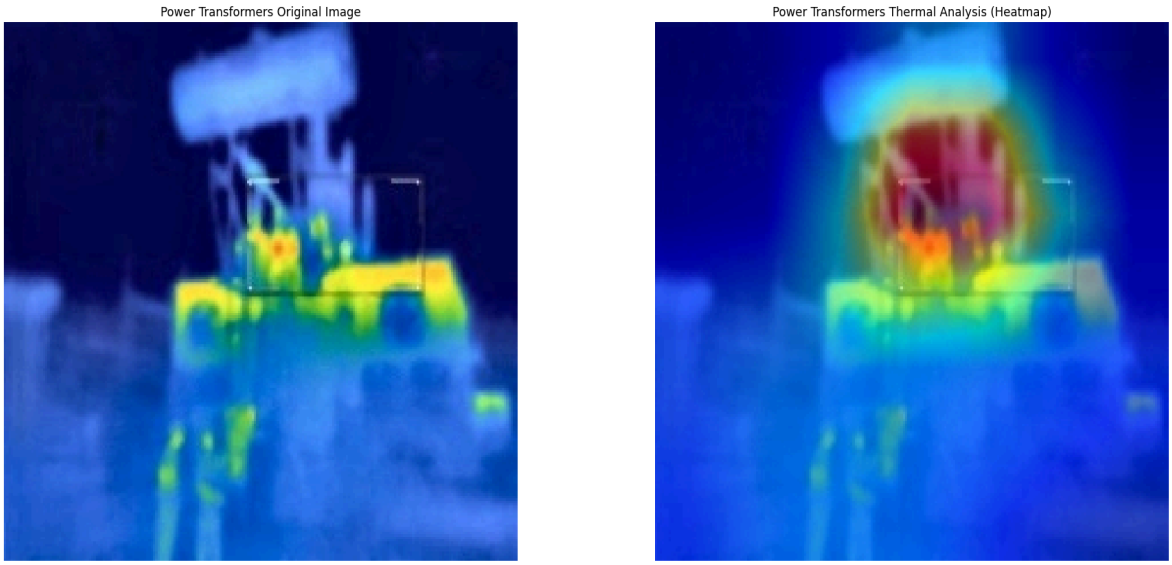

```
Equipment: Surge Arrestors
Condition: Abnormal
Hotspot Location: upper section
Fault Probability: 0.968

AI Explanation:
 The hotspot is located in the upper section of the surge arrester. The
surge arrestor was installed to protect the conductive parts from
excessive voltage fluctuations caused by high-voltage power lines. When
the current draw exceeds its capacity, the heat generated causes thermal
damage to the equipment. This can result in severe injuries or even
death if not handled appropriately. It's essential that the device be
tested periodically for proper operation and maintenance, and any signs
of abnormality should be reported immediately.
```

Fig 10 - Grad-CAM and SHAP-Based Interpretation of Abnormal Condition in Surge Arrestors with AI-based explanation

The high fault probability shown in Fig 10 indicates a severe operating condition, potentially associated with excessive voltage stress or insulation degradation within the surge arrester.

5.4 Normal Condition - Power Transformers

The system was put to the test using a thermal image of a power transformer, shown in Figure 11. The model had no hesitation in classifying it as normal, giving a fault probability of just 0.1. The Grad-CAM heatmap backs this up, showing heat spread evenly across the mid-section with nothing unusual standing out. The extracted features paint the same picture, reflecting steady thermal behavior that clearly points to a healthy, well-functioning transformer.

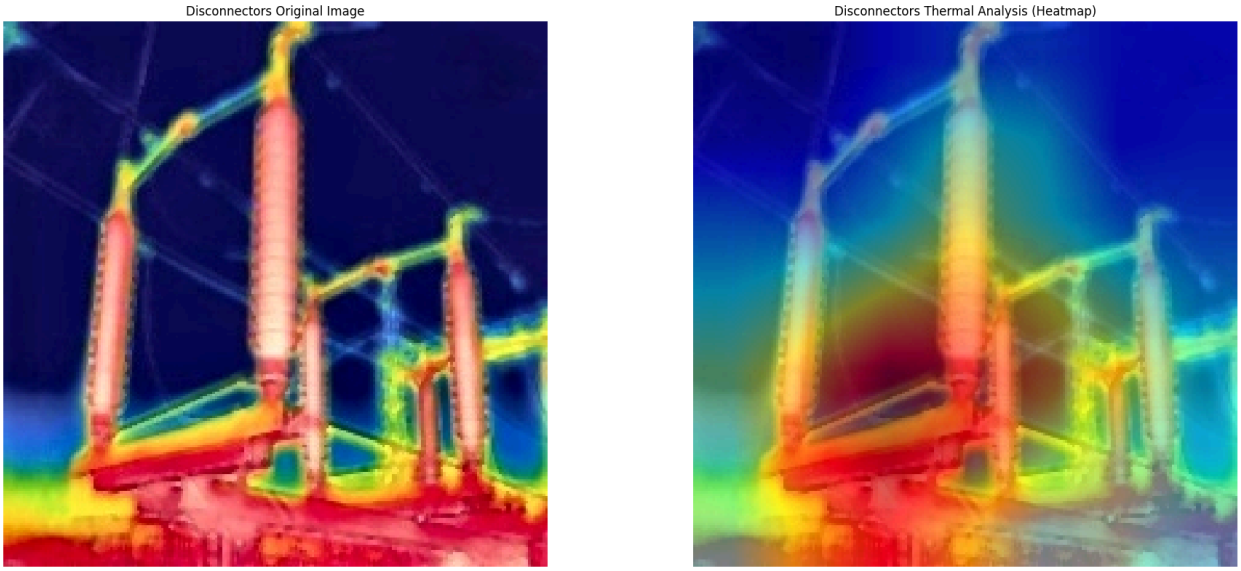



```
Equipment: Power Transformers
Condition: Normal
Hotspot Location: middle section
Fault Probability: 0.1

AI Explanation:
 During normal operating conditions, transformer temperature is
typically around 105-125°C (231-261°F) with no signs of overheating or
failure.
```

Fig 11 - Grad-CAM and SHAP-Based Interpretation of Normal Condition in Power Transformer with AI-based explanation

5.5 Abnormal Condition - Disconnectors

The evaluation of the proposed system made use of a thermal image of a disconnector, presented in Fig 12. This visualization integrates the original image, Grad-CAM heatmap, and generated explanation. The model classified the condition as abnormal, with a fault probability of 0.9.




```
Equipment: Disconnectors
Condition: Abnormal
Hotspot Location: lower section
Fault Probability: 0.9

AI Explanation:
 Electricity enters the circuit through terminals A-C (hot). The
disconnector is designed to protect the conductor from short circuits
caused by thermal expansion in hotter sections. If there is no current
flowing through this section, it will not heat up enough to trigger the
disconnection mechanism, which means that the disconnected segments
remain connected. This causes excessive resistance in the circuit and
can lead to overheating of other parts, such as transformers or motors,
leading to failure due to spark ignition. Additionally, since the
disconnector has a low switching frequency, it may fail to detect any
potential faults, thus increasing the likelihood of catastrophic
breakdown. Therefore, proper maintenance and inspection should be
conducted regularly to ensure optimal performance and prevent
unnecessary risks.
```

Fig 12 - Grad-CAM and SHAP-Based Interpretation of Abnormal Condition in Disconnector with AI-based explanation

The Grad-CAM highlights a hotspot in the lower section, indicating localized thermal concentration. This suggests abnormal heat accumulation, which may be caused by overheating or electrical stress. Overall, the results confirm that the hybrid model achieves superior performance in thermal fault detection. The use of XAI ensures interpretability, while TinyLLaMA provides meaningful explanations, making the system accurate, reliable, and suitable for practical applications.